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Percentage

प्रतिशत

Conceptually Imp. for all chapters of Arithmetic

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Per cent

Per → With respect to / Compared to

Cent → 100

प्रति → w.r.t.

शत → 100

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Percent : A value when checked w.r.t. 100

30%

100

Three Tools : (Comparision)

1.) Difference

2.) Ratio

3.) Percentage

A person spends 20% of his salary.

$20\% = \frac{20}{100}$

$= \frac{1}{5}$

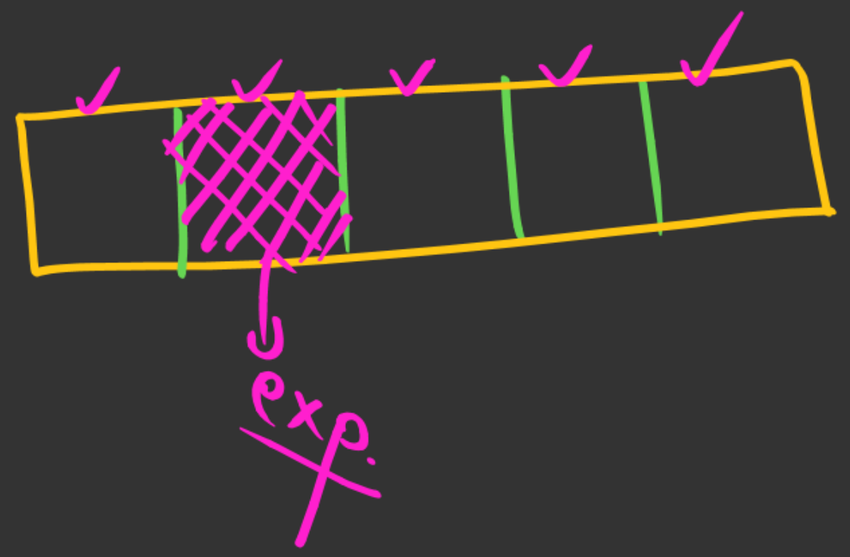
20% → $\frac{20 \text{ out of}}{100}$ [100 mein se 20]

$\frac{\cancel{20}}{\cancel{100}}$ $\frac{1}{5}$

$\frac{1}{5}$

Salary = 100

Expense = 20



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$$25\% = \frac{25}{100} = \frac{1}{4}$$

If a total of 100 parts are there, we are talking about 25 out of them.

$$25\% = \frac{1}{4}$$

Hota Hai

ka

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Ratio: Ratio is a comparision by division

B_1	B_2
20	30

$$\frac{B_1}{B_2} = \frac{2\cancel{0}}{3\cancel{0}} = \left(\frac{2}{3}\right)$$

$$\frac{B_1}{B_2} = \frac{2}{3}$$

$B_1 : B_2$

$2 : 3$

Ratio should be solved to its lowest terms

B_1	B_2
~~20~~	~~30~~
10	15

$$\frac{B_1}{B_2} = \frac{\cancel{10}\ 2}{\cancel{15}\ 3}$$

$B_1 : B_2$

$10 : 15$

$2 : 3$

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Percentage is also a ratio compared to 100.

B	G	Total
40	60	100

Ratio:

$$\frac{B}{G} = \frac{40}{60} = \frac{2}{3}$$

$$\boxed{\begin{matrix} B : G \\ 2 : 3 \end{matrix}}$$

B : G

2 : 3

Boys → 2 parts

Girls → 3 parts

Total students → 5 parts

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Boys contribute to what portion of class

$$\frac{2}{5}$$

2 → Boys

5 → whole class

$$\frac{40}{100} \times 100$$

$$\Rightarrow 40\% = \frac{2}{5}$$

Girls contribute to what portion of class.

$$\frac{3}{5}$$

3 — Girls

5 — Whole class

$$\frac{60}{100} \times 100$$

$$60\% = \frac{3}{5}$$

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Chetan

160 — obtained

400 — max

$$\frac{160}{400} \times 100$$

40 %

Surbhi

180 — obtained

600 — max

$$\frac{180}{600} \times 100$$

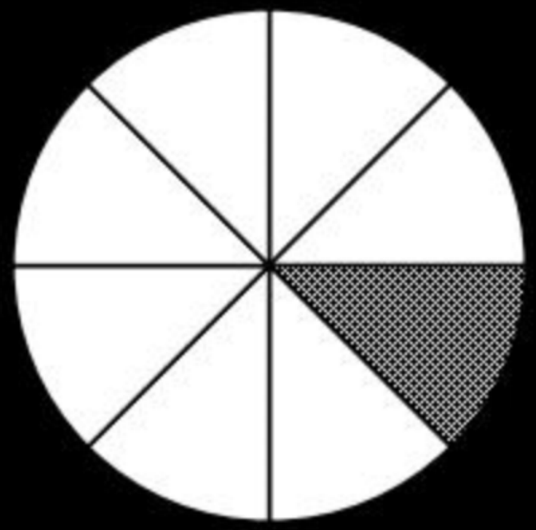
30 %

$\frac{1}{8} \times 100$

$\Rightarrow 12.5\%$

$\frac{1}{5} \times 100 = 20\%$

1 in 8



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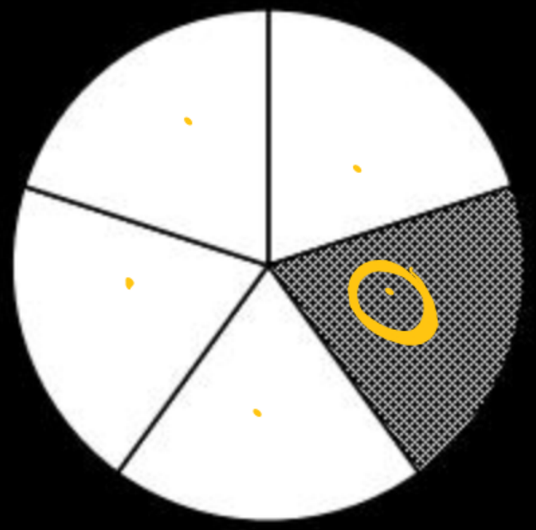


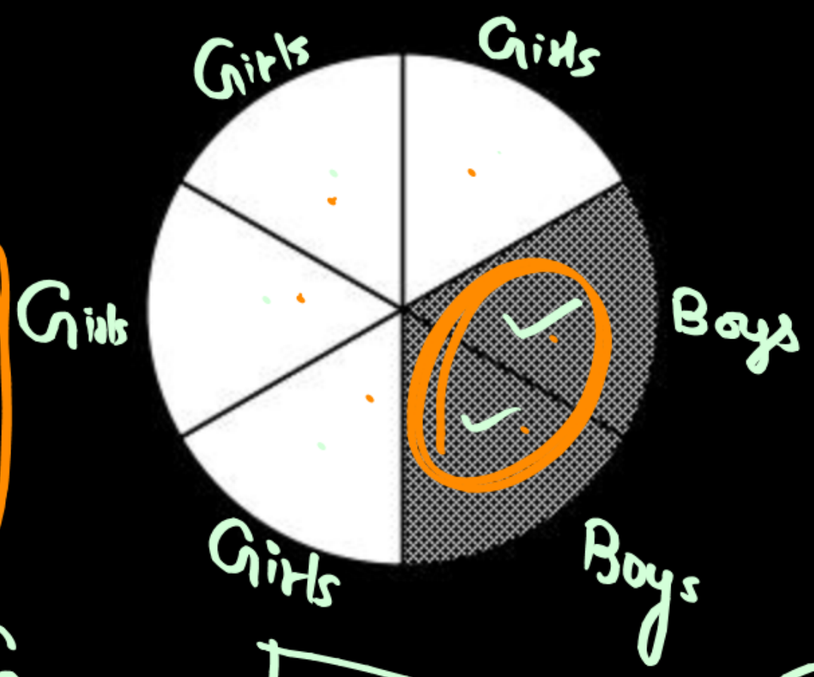
1 in 5

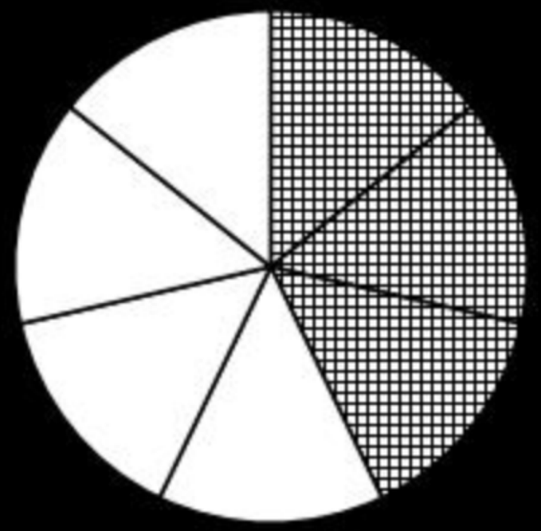
$\frac{2 \rightarrow B}{6 \rightarrow G}$

$\frac{\cancel{2}^{1}}{\cancel{6}_{3}} \times 100$

$\boxed{33.33\% \hookrightarrow \text{Boys}}$

Total Students





B	G
$\cancel{2}$	$\cancel{4}$

1 : 2

$\Rightarrow \boxed{B : G \quad 1 : 2} \Rightarrow \frac{B}{Total} = \frac{1}{3}$

$\frac{G}{Total} = \frac{2}{3}$

Amit obtained 30 marks out of 75 Marks, find his Marks percent.

75 → 30

1 → $\frac{30}{75}$

100 → $\frac{30}{75} \times 100$

Here we are taking total Marks as 100

$\frac{30}{75}$

$\frac{30}{75} \times 100$

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$$\left(\frac{200}{500}\right) \times 100 \Rightarrow 40\%$$

$$\left(\frac{2}{5}\right) = 40\%$$

$$1500 \longrightarrow 225$$

$$1 \longrightarrow \frac{225}{1500}$$

$$100 \longrightarrow \frac{225}{1500} \times 100$$

$$\Downarrow$$

$$15\%$$

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Chintu

1500

exp. = 225

1500 → 225

1 → $\frac{225}{1500}$

100 → $\frac{225}{1500} \times 100$

⇒ 15%

Mintu

1800

270 (exp.)

1800 → 270

1 → $\frac{270}{1800}$

100 → $\frac{270}{1800} \times 100$

⇒ 15%

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$15\% = \frac{15}{100}$

$25\% = \frac{25}{100}$

$80\% = \frac{80}{100}$

$75\% = \frac{75}{100}$

$30\% = \frac{30}{100}$

$35\% = \frac{35}{100}$

$45\% = \frac{45}{100}$

8 ← value whose % is being calc.

40 ⟹ Total value

$\frac{8}{40} \times 100 \Rightarrow 20\%$

$\frac{8}{50} \times 100 \Rightarrow 16\%$

$\frac{16}{80} \times 100 \Rightarrow 25\%$

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$$24\% = (4\%) \times 6$$

$$= \frac{1}{25} \times 6$$

$$\boxed{24\% = \frac{6}{25}}$$

$$24\% = \left(\frac{\cancel{24}}{\cancel{100}}\right)\frac{6}{25} = \frac{6}{25}$$

$$24\% = \frac{6}{25}$$

ka

Base Value

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$37.5\% \Rightarrow 12.5\% \times 3$

$\Rightarrow \frac{1}{8} \times 3$

$37.5\% = 3$ ka 8

$48\% = 4\% \times 12$

$= \frac{1}{25} \times 12$

$48\% = 12$ ka 25

$\frac{48}{100}$ 12 25

$60\% = 20\% \times 3$

$60\% = \frac{3}{5}$ ka

$62.5\% = 12.5\% \times 5$

$= \frac{1}{8} \times 5$

$62.5\% = 5$ 8

$22.22\% = (11.11\% \times 2)$

$= \frac{1}{9} \times 2$

$22.22\% = 2$ ka 9

$36.36\% = 9.09\% \times 4$

$= \frac{1}{11} \times 4$

$36.36\% = 4$ 11

$45\% = 5\% \times 9$

$= \frac{1}{20} \times 9$

$45\% = 9$ ka 20

$65\% = 5\% \times 13$

$= \frac{1}{20} \times 13$

$65\% = 13$ ka 20

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$325\,|\cdot = 300\,|\cdot + 25\,|\cdot$

$= 3 + \frac{1}{4} \Rightarrow \frac{13}{4}$

$152\,|\cdot = 100\,|\cdot + 52\,|\cdot$

$1 + \frac{13}{25} = \frac{38}{25}$

$502\,|\cdot = 500\,|\cdot + 2\,|\cdot$

$= 5 + \frac{1}{50}$

$= \frac{251}{50}$

$412.5\,|\cdot = 400\,|\cdot + 12.5\,|\cdot$

$= 4 + \frac{1}{8}$

$= \frac{33}{8}$

$225\,|\cdot \Rightarrow 25\,|\cdot \times 9$

$\Rightarrow 9/4$

$277.77\,|\cdot = 200\,|\cdot + 77.77\,|\cdot$

$= 2 + 7/9$

$= 25/9$

$416.66\,|\cdot = 400\,|\cdot + 16.66\,|\cdot$

$= 4 + 1/6$

$= 25/6$

$100\,|\cdot = 1$

$200\,|\cdot = 2$

$300\,|\cdot = 3$

$400\,|\cdot = 4$

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$27.77\dot{1}$

$20\dot{1} + 7.77\dot{1}$

$\frac{1}{5} + \frac{7}{90}$

$\frac{18+7}{90} \Rightarrow \frac{\cancel{25}\ 5}{\cancel{90}\ 18}$

$77.77\dot{1} = \frac{7}{9}$

$7.77\dot{1} = \left(\frac{7}{90}\right)$

$27.77\dot{1} = \frac{5}{18}$

$\boxed{5.55\dot{1} \times 5}$

$\frac{1}{18} \times 5$

$\frac{5}{18}$

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Percentage Increase/Decrease :

If a Number is increased by 20 %

20 % = $\frac{1}{5}$ ka

Number = 100

Increase = +20

New = 120

Initial : New

100 : 120 ⇒ 5 : 6

Initial → final

5 $\xrightarrow{+1}$ 6

5 : 6

If a number is increased by 50 %

50 % = $\frac{1}{2}$ ka

Number = 100

Increase = +50

New = 150

Initial : final

2 $\xrightarrow{+1}$ 3

2 : 3

Initial : New

~~100~~ : ~~150~~

2 : 3

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If a number Increases by 37.5%

37.5% = 3/8

ka

Number = 100

+ Increase +37.5

New = 137.5

Initial : final

8 11

+3

Initial New

100 137.5

8 : 11

If a number is increased by 36.36%

36.36% = 4/11

ka

Number = 100

+ Inc. +36.36

New = 136.36

Initial : final

11 : 15

+4

Initial New

100 136.36

11 : 15

What is 24% of 2400?

24% of 2400 ⇒ 576

24% = $\left(\frac{6}{25}\right)$

$\frac{6}{25} \times 2400$

What is 16.67% of 1680?

16.67% = $\frac{1}{6}$

$\frac{1}{6} \times 1680$

24 is what percent of 80?

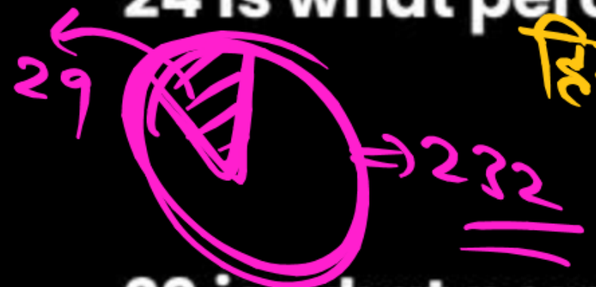
हिस्सा

$\frac{24}{80}$ | 3 / 10

$\frac{3}{10} \times 100$

⇒ 30%

29 is what percent of 232?



$\frac{29}{232}$ ⇒ $\frac{1}{8} \times 100$

⇒ 12.5%